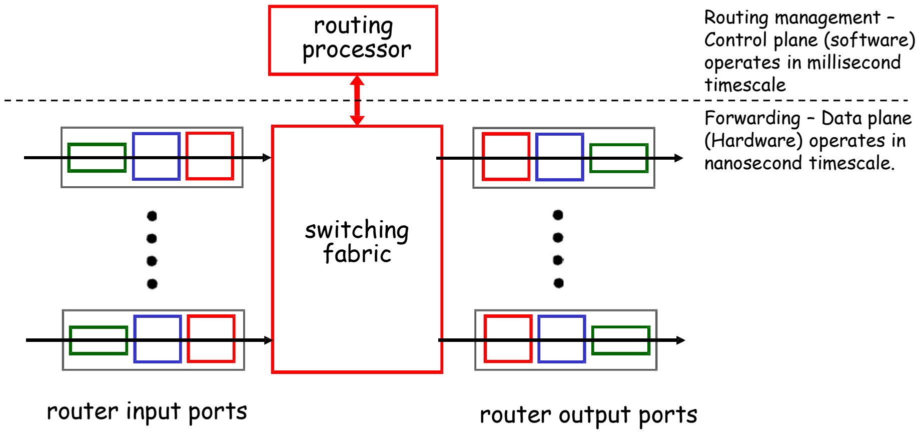
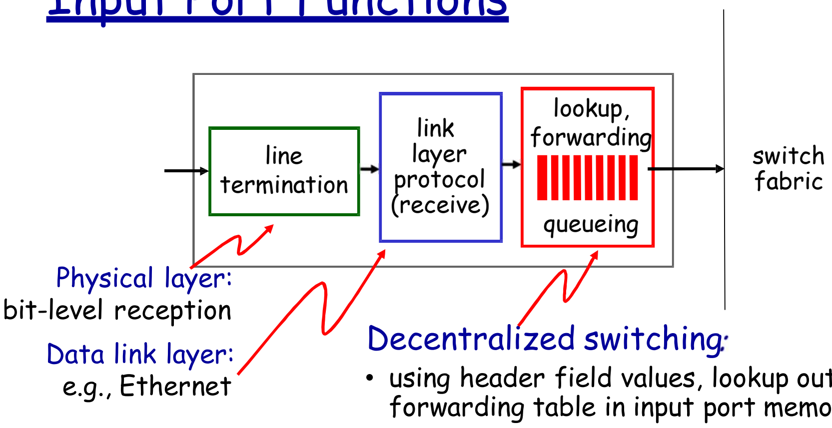
Router Architecture

High-level view of generic router architecture with two key router functionalities:

- run routing algorithms/protocol
- forwarding datagrams from incoming to outgoing link



Input Port Functions

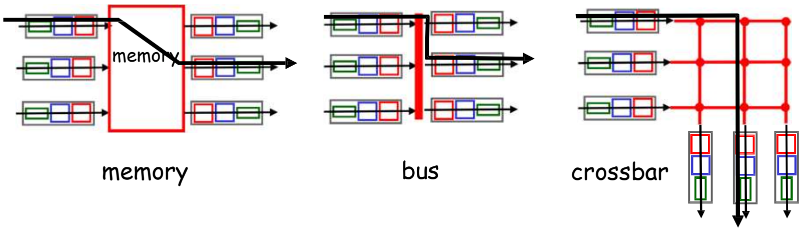


Decentralized switching:

- using header field values, lookup output port using forwarding table in input port memory
- goal: complete input port processing at
- line speed
- queuing: if datagrams arrive faster than forwarding rate into switch fabric
- destination-based forwarding: forward based only on destination IP address (traditional)
- generalized forwarding: forward based on any set of header field values

Switching fabrics

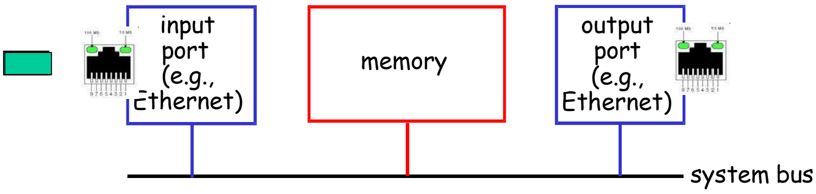
- transfer packet from input buffer to appropriate output buffer
- switching rate: rate at which packets can be transfer from inputs to outputs
  - often measured as multiple of input/output line rate
  - N inputs: switching rate N times line rate desirable
- three types of switching fabrics



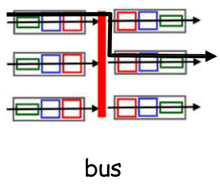
Switching Via Memory

First generation routers:

- traditional computers with switching under direct control of CPU
- packet copied to system's memory
- speed limited by memory bandwidth (2 bus crossings per datagram)

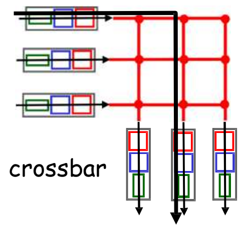


Switching Via a Bus



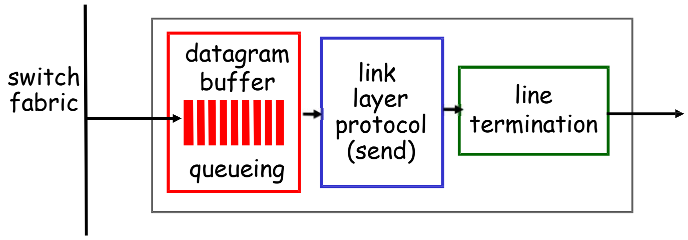
- datagram from input port memory to output port memory via a shared bus
- bus contention: switching speed limited by bus bandwidth
- 32 Gbps bus, Cisco 5600: sufficient speed for access and enterprise routers

Switching Via An Interconnection Network



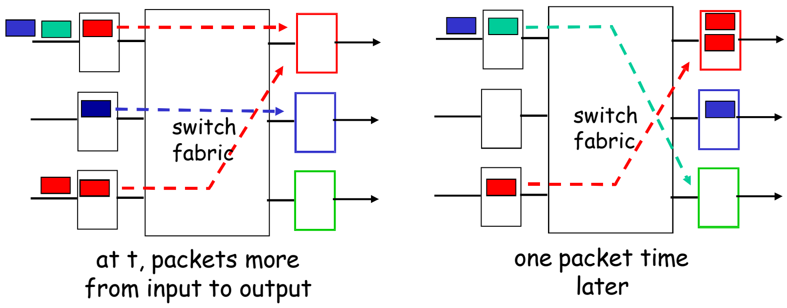
- overcome bus bandwidth limitations
- Banyan networks, crossbar, other interconnection nets initially developed to connect processors in multiprocessor
- advanced design: fragmenting datagram into fixed length cells, switch cells through the fabric.
- Cisco 12000: switches 60 Gbps through the interconnection network

Output Ports



- *buffering* required when datagrams arrive from fabric faster than the transmission rate
- *scheduling discipline* chooses among queued datagrams for transmission

Output port queueing



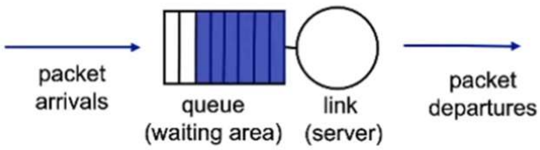
- buffering when arrival rate via switch exceeds output line speed
- *queueing (delay) and loss due to output port buffer overflow!*

How much buffering?

- RFC 3439 rule of thumb: average buffering equal to typical RTT (say 250 msec) times link capacity C
  - e.g., C = 10 Gpbs link: 2.5 Gbit buffer
- recent recommendation: with N flows, buffering equal to $\frac{RTT \cdot C}{\sqrt{N}}$

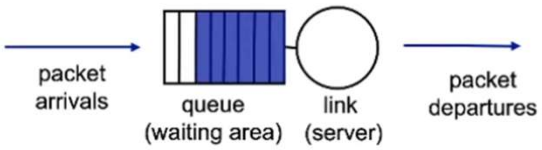
How to buffer?

- Dropping – choose next packet to discard
- Discard policy – if packet arrives to full queue, who to discard?
  - > tail drop: drop arriving packet
  - > priority: drop/remove on priority basis
  - > random: drop/remove randomly



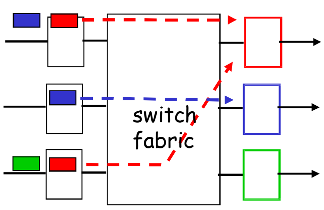
How to de-buffer?

- Scheduling – choose next packet to send on link
- Scheduling policy -
  > FIFO scheduling: first in first out, send in order of arrival queue
  > Class-based scheduling: divide packets into classes
    - priority: certain class gets to go first
    - RR (Round-robin): rotating turns
    - WFQ (weighted fair queuing): weighted rotating turns

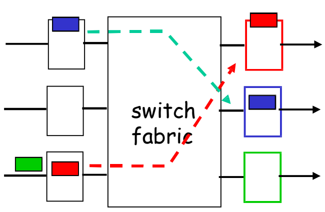


Input Port Queuing

- fabric slower than input ports combined -> queueing may occur at input queues
  - *queueing delay and loss due to input buffer overflow!*
- Head-of-the-Line (HOL) blocking: queued datagram at front of queue prevents others in queue from moving forward



output port contention: only one red datagram can be transferred. *lower red packet is blocked*



one packet time later: green packet experiences HOL blocking